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PROJECT REPORT

TITLE : Library Management System

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ABSTRACT

The Library Management System is a Python-based application developed to manage and organize books in a library efficiently. The main objective of this project is to provide a simple system for storing, searching, issuing, and returning books.

This application allows users to add new books with details such as title and author, view the list of available books, and search for specific books. It also includes functionality to issue and return books by updating their availability status. Additionally, users can delete book records when required.

The system is implemented using basic Python concepts such as lists, dictionaries, functions, loops, and conditional statements. It demonstrates how data can be stored and manipulated effectively in a real-world scenario.

Overall, the Library Management System helps in maintaining book records systematically and reduces manual effort, making library operations more efficient and organized.

SOURCE CODE:

```python
# Library Management System

library = []


def add_book():
    title = input("Enter book title: ")
    author = input("Enter author name: ")

    book = {
        "title": title,
        "author": author,
        "issued": False
    }

    library.append(book)
    print("✅ Book added successfully!")


# ---------------- VIEW BOOKS ----------------
def view_books():
    if not library:
        print("❌ No books available")
        return

    print("\n📚 Book List:")
    for i, book in enumerate(library):
        status = "Issued" if book["issued"] else "Available"
        print(f"{i+1}. {book['title']} by {book['author']} - {status}")


# ---------------- SEARCH BOOK ----------------
def search_book():
    name = input("Enter book title to search: ")
    found = False

    for book in library:
```

```python
        if name.lower() in book["title"].lower():
            print(book)
            found = True

    if not found:
        print("❌ Book not found")


# ---------------- ISSUE BOOK ----------------
def issue_book():
    view_books()
    index = int(input("Enter book number to issue: ")) - 1

    if 0 <= index < len(library):
        if not library[index]["issued"]:
            library[index]["issued"] = True
            print("📕 Book issued successfully!")
        else:
            print("❌ Book already issued")
    else:
        print("❌ Invalid choice")


# ---------------- RETURN BOOK ----------------
def return_book():
    view_books()
    index = int(input("Enter book number to return: ")) - 1

    if 0 <= index < len(library):
        if library[index]["issued"]:
            library[index]["issued"] = False
            print("📗 Book returned successfully!")
        else:
            print("❌ Book was not issued")
    else:
        print("❌ Invalid choice")
```

```python
# ---------------- DELETE BOOK ----------------
def delete_book():
    view_books()
    index = int(input("Enter book number to delete: ")) - 1

    if 0 <= index < len(library):
        library.pop(index)
        print("🗑️ Book deleted!")
    else:
        print("❌ Invalid choice")


# ---------------- MENU ----------------
while True:
    print("\n===== LIBRARY MENU =====")
    print("1. Add Book")
    print("2. View Books")
    print("3. Search Book")
    print("4. Issue Book")
    print("5. Return Book")
    print("6. Delete Book")
    print("7. Exit")

    choice = input("Enter your choice: ")

    if choice == '1':
        add_book()
    elif choice == '2':
        view_books()
    elif choice == '3':
        search_book()
    elif choice == '4':
        issue_book()
    elif choice == '5':
        return_book()
    elif choice == '6':
        delete_book()
```

```python
    elif choice == '6':
        delete_book()
    elif choice == '7':
        print("Exiting...")
        break
    else:
        print("❌ Invalid choice")
```

```python
        update_contact()
    elif choice == '5':
        delete_contact()
    else:
        break
```

Output:

```
===== LIBRARY MENU =====
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Delete Book
7. Exit
Enter your choice: 1
Enter book title: Django basics
Enter author name: Python
✅ Book added successfully!

===== LIBRARY MENU =====
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Delete Book
7. Exit
Enter your choice: 1
Enter book title: Python
Enter author name: Guido van rossum
✅ Book added successfully!

===== LIBRARY MENU =====
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Delete Book
7. Exit
Enter your choice: 2

📚 Book List:
1. Django basics by Python - Available
2. Python by Guido van rossum - Available

===== LIBRARY MENU =====
```

```
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Delete Book
7. Exit
Enter your choice: 3
Enter book title to search: Python
{'title': 'Python', 'author': 'Guido van rossum', 'issued': False}

===== LIBRARY MENU =====
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Delete Book
7. Exit
Enter your choice: 6

📚 Book List:
1. Django basics by Python - Available
2. Python by Guido van rossum - Available
Enter book number to delete: 1
🗑Book deleted!

===== LIBRARY MENU =====
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Delete Book
7. Exit
Enter your choice: 7
Exiting...
PS C:\Users\reeth\OneDrive\Desktop\New folder>
```